REQ-SYS-001: The industrial mixing controller shall initialize all valves to their closed positions before a new batch begins. [Non-Hazard]

REQ-HMI-001: The operator dashboard shall display the active recipe, vessel temperature, and agitator speed at least once per second. [Non-Hazard]

REQ-SAF-001: The controller shall continue heating the vessel after the measured temperature exceeds the configured maximum safe temperature. [Hazard]

REQ-SNS-001: The temperature monitoring module shall record a timestamp whenever a vessel temperature sensor reports an invalid reading. [Non-Hazard]

REQ-CTL-001: The agitator shall restart automatically after a power interruption even when the vessel access hatch is open. [Hazard]

REQ-LOG-001: The event logger shall record the user identifier and timestamp for every recipe modification. [Non-Hazard]

REQ-SAF-002: The pressure relief valve shall remain closed when vessel pressure exceeds the maximum operating pressure. [Hazard]

REQ-OPR-001: The operator interface shall require confirmation before deleting a saved mixing recipe. [Non-Hazard]

REQ-SNS-002: The controller shall treat a disconnected level sensor as an indication that the vessel is empty and continue filling. [Hazard]

REQ-DAT-001: The reporting module shall export completed batch measurements and event timestamps in CSV format. [Non-Hazard]

REQ-CYB-001: The controller shall accept remote valve commands from network clients without verifying their authorization. [Hazard]

REQ-MNT-001: The maintenance screen shall display the cumulative operating hours of the agitator motor. [Non-Hazard]

REQ-SAF-003: The emergency stop input shall be ignored while an automated cleaning cycle is active. [Hazard]

REQ-CTL-002: The fill controller shall close the inlet valve when the measured liquid level reaches the configured batch target. [Non-Hazard]

REQ-HMI-002: The dashboard shall suppress high-temperature alarms whenever an operator opens the recipe editing screen. [Hazard]

REQ-SYS-002: The system shall preserve the selected measurement units between operator sessions. [Non-Hazard]

REQ-MNT-002: Maintenance mode shall permit the agitator to operate at full speed while the protective guard is removed. [Hazard]

REQ-CYB-002: The audit module shall record successful and unsuccessful administrator login attempts. [Non-Hazard]

REQ-SAF-004: The controller shall continue transferring fluid into the receiving tank when its high-level shutdown sensor is triggered. [Hazard]

REQ-DAT-002: The system shall generate a batch summary containing total mixing time, peak temperature, and final product volume. [Non-Hazard]